



Anton Kundenko

Principal Systems Engineer • Architect of High-Performance Database Systems • Programming Language Creator

18+ years in performance-critical systems — mostly designing original database engines and the programming languages that drive them. Creator of **ThePlatform** and **RayforceDB**, and the languages **O** and **Rayfall**.

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◆ IEEE Senior Member · 2026

SUMMARY

Principal systems engineer and programming language creator with **18+ years** of experience building performance-critical infrastructure. Created two original database systems — **RayforceDB** and **ThePlatform** — each with its own purpose-built programming language (**Rayfall** and **O**), now serving as core frameworks across **hundreds of production instances** at **Lynx Trading Technologies**. Because no database on the market offered the same combination of capabilities, these systems let Lynx open **entirely new business areas** — advanced pre-trade risk management, locate routing (a marketplace for borrowed shares), pre-trade compliance, and rapid entry into derivatives markets, the most technologically demanding sector in finance — plus client-scale reporting that would otherwise be impossible to run profitably. Today they power Lynx's most critical operations: real-time market-data ingestion, risk, monitoring, order-routing analytics, and regulatory reporting. Deep expertise in SIMD vectorization, columnar storage design, language runtime implementation, and low-latency systems architecture.

ORIGINAL CONTRIBUTIONS

RayforceDB — Creator & Sole Architect

2020 — Present · rayforcedb.com

Original SIMD-first columnar analytical database (<1MB binary, zero dependencies) with **Rayfall** — an original Lisp-like array language for vectorized analytics. Created independently; subsequently sponsored by Lynx Trading Technologies for production adoption.

- Designed and implemented a novel database architecture from scratch in pure C: SIMD-vectorized operators (AVX2/NEON), cache-aware columnar layouts, mmap-based storage, and custom memory management.
- Created the Rayfall language: a composable, Lisp-like query language with first-class vectors, tables, dictionaries, temporal types, and a comprehensive type system — offering an alternative paradigm to SQL and kdb+'s q for analytical workloads.
- Built a complete ecosystem: core engine, Python bindings (DataFrame API), Rust bindings, WebAssembly runtime, VS Code extension with REPL, benchmarking suite, and documentation.
- Adopted by Lynx Trading for production financial analytics — Lynx sponsored continued development based on the system's performance advantages over incumbent solutions.
- Independently adopted in production beyond its origin at Lynx — **Fortitudo Technologies** runs RayforceDB for time-series analytics, validating the system outside its sponsoring firm.

Rayfall language

Pure C

SIMD (AVX2/NEON)

Columnar

<1MB / zero deps

MIT licensed

Cross-platform

ThePlatform — Creator & Lead Architect

2016 — Present · theplatform.technology

Columnar time-series database and real-time analytics stack created for **Lynx Trading Technologies**, featuring the original **O programming language** — a vector language inspired by APL/k with first-class tables, async reagents, and join-calculus concurrency.

- Designed and implemented a complete database system with its own programming language, replacing Lynx's kdb+/q infrastructure for real-time market data, risk analytics, and order flow processing.
- Created the O language: vector-oriented with nanosecond timestamps, streaming pub/sub, pattern matching, and declarative concurrency — purpose-built for financial time-series workloads.
- Built async communication reagents (TCP/UDP/WebSocket/IPC/TLS), plugin architecture, and standard library enabling Lynx to consolidate multiple vendor tools into one platform.
- Deployed across hundreds of production instances as a core framework for Lynx's most critical operations — market-data ingestion, risk, monitoring, order-routing analytics, and regulatory reporting.

O language

Columnar DB

Time-series

Rust

Real-time streaming

kdb+ replacement

FOCUS

SPECIALIZATION

Database systems architecture • Programming language design • SIMD-vectorized execution engines

DOMAINS

Financial analytics • Real-time trading systems • Time-series • Embedded analytics

APPROACH

Original from-scratch designs with zero dependencies, measurable performance, and production-grade reliability

RECOGNITION

PROFESSIONAL MEMBERSHIP
IEEE SENIOR MEMBER

Spain Section · Elected 2026
Member #102451070 · Valid through Dec 2026

TECHNICAL SKILLS

ORIGINAL LANGUAGES CREATED

Rayfall

O language

IMPLEMENTATION LANGUAGES

C (C17)

Rust

Python

k/q (kdb+)

Lisp/Scheme

PERFORMANCE

SIMD (AVX2/NEON)

Profiling (perf)

Cache optimization

Zero-copy

Memory layouts

DATABASES

Columnar storage

Vectorized execution

Query runtimes

Index structures

Time-series

SYSTEMS

Linux

ARM

mmap I/O

Concurrency

Lock-free

TOOLING

Git

Make/CMake

Sanitizers

Benchmarks

CI

INDUSTRY ADOPTION

TeideDB — Creator & Lead Architect

2024 — Present · teidedb.com

Analytical SQL engine / embedded analytics runtime extending the Rayforce ecosystem with a modern SQL interface over a vectorized, in-process execution model.

- Vectorized operators and optimizer-oriented architecture for fast analytical workloads.
- Designed for agent-driven analytics pipelines and interactive reporting over columnar data.

SQL	Vectorized execution	Embedded / in-process	C
• Easy to learn • Easy to use • Easy to integrate • Easy to extend • Easy to maintain • Easy to debug • Easy to test • Easy to deploy • Easy to scale • Easy to secure • Easy to monitor • Easy to manage • Easy to upgrade • Easy to migrate • Easy to backup • Easy to restore • Easy to recover • Easy to backup • Easy to restore • Easy to recover	• Fast • Efficient • Scalable • Secure • Reliable • Available • Portable • Extensible • Configurable • Customizable • Flexible • Adaptable • Resilient • Robust • Stable • Consistent • Accurate • Precise • Detailed • Comprehensive • Thorough • Exhaustive • In-depth • Extensive • Broad • Wide • Deep • Far-reaching • Comprehensive • Thorough • Exhaustive • In-depth • Extensive • Broad • Wide • Deep • Far-reaching	• Fast • Efficient • Scalable • Secure • Reliable • Available • Portable • Extensible • Configurable • Customizable • Flexible • Adaptable • Resilient • Robust • Stable • Consistent • Accurate • Precise • Detailed • Comprehensive • Thorough • Exhaustive • In-depth • Extensive • Broad • Wide • Deep • Far-reaching • Comprehensive • Thorough • Exhaustive • In-depth • Extensive • Broad • Wide • Deep • Far-reaching	• Fast • Efficient • Scalable • Secure • Reliable • Available • Portable • Extensible • Configurable • Customizable • Flexible • Adaptable • Resilient • Robust • Stable • Consistent • Accurate • Precise • Detailed • Comprehensive • Thorough • Exhaustive • In-depth • Extensive • Broad • Wide • Deep • Far-reaching • Comprehensive • Thorough • Exhaustive • In-depth • Extensive • Broad • Wide • Deep • Far-reaching

Teidelum — Creator & Lead Architect

2025 — Present · lum.teidedb.com

Open-source knowledge infrastructure built on the Teide columnar engine — syncs work tools, indexes content, and serves data to AI agents via MCP and applications via REST.

- BM25-ranked full-text search, SQL analytics, graph traversal; dual API (MCP + REST).
- Single Rust binary, zero cloud dependencies, incremental sync from Notion/Zulip/databases.

Rust MCP Full-text search SQL MIT licensed

AXL DB2021 — Present · axl-db.com

Ultra-compact (<1MB core) zero-dependency vector database for deterministic-latency workloads (HFT, edge analytics, embedded ARM/x86).

Vector DB <1MB core AVX2/NEON

hyperbridge

Open Source · [GitHub](#)

Lock-free multi-producer/multi-consumer unbounded channel with async support in Rust.

Rust Lock-free Concurrency

EXPERIENCE

Principal Systems Engineer & Architect of High-Performance Database Systems

2013 — Present

Lynx Trading Technologies

- Created **ThePlatform** (2016) — a columnar time-series database and the **O programming language**, replacing Lynx's kdb+/q infrastructure for real-time market data, risk analytics, and order flow processing.
- Created **RayforceDB** (2020) — an original SIMD-first columnar database and the **Rayfall language**; independently developed, then sponsored by Lynx for production adoption and continued development.
- Enabled **entirely new lines of business** impossible on existing databases — advanced pre-trade risk management, locate routing (a marketplace for borrowed shares), pre-trade compliance, derivatives trading, and client-scale regulatory reporting — across **hundreds of production instances**.
- Built low-latency trading infrastructure, market-data processing pipelines, and real-time analytics systems for US equity markets.

Lead Engineer, Project Lead

2012 — 2013

Samsung Electronics Ukraine R&D

- Led OS virtualization work on ARM Cortex-A15; implemented TrustZone security extensions and DRM systems.
- Delivered low-level platform software integrating kernel, security, and hardware-software boundaries.

Embedded / Systems Developer

2007 — 2012

TranSat • Technotrade

- Built embedded systems (smart house, voting, CCTV, parking control) and POS/payment terminal software.
- Developed transaction-critical systems with strong reliability requirements.

EDUCATION

Specialist Degree (Master-level) — Automation & Computer-Integrated Technologies

Donbass State Engineering Academy • 2001 — 2006

PRODUCTION USE

Core frameworks across hundreds of production instances at Lynx Trading Technologies — market-data ingestion, risk, monitoring, order-routing analytics, and regulatory reporting

NEW CAPABILITIES

Enabled derivatives trading, pre-trade risk & compliance, locate routing, and client-scale reporting — business areas no existing database could support

INDEPENDENT ADOPTION

RayforceDB also runs in production beyond Lynx — independently adopted by [Fortitudo Technologies](#) for time-series analytics

OPEN SOURCE

RayforceDB, TeideDB, Teidelum, hyperbridge
— MIT licensed
github.com/singaraiona

Automation engineering foundation: embedded systems, computer-integrated technologies, and systems programming.